Fourier series representations

We already looked at series when we talked about how to find the series solution to a second order differential equation, so we're familiar with representing a function (and therefore the curve of a general solution) as a series.

But so far we've only represented functions as Taylor series, and that series type has limited usefulness. For example, to use a Taylor series to represent a function $f(x)$, the function has to be sufficiently differentiable, because we need its derivatives for the coefficients in the series representation,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

We can also run into an issue if the radius of convergence R, given by $|x-a| < R$, is too small to make the series useful.

So instead of using a Taylor series, sometimes it'll be better (or necessary) to use a different kind of series representation.

Fourier series representation

A **Fourier series** is a different kind of series representation. Instead of using powers of $(x-a)$, like a Taylor series, it uses sine and cosine functions of different arguments.

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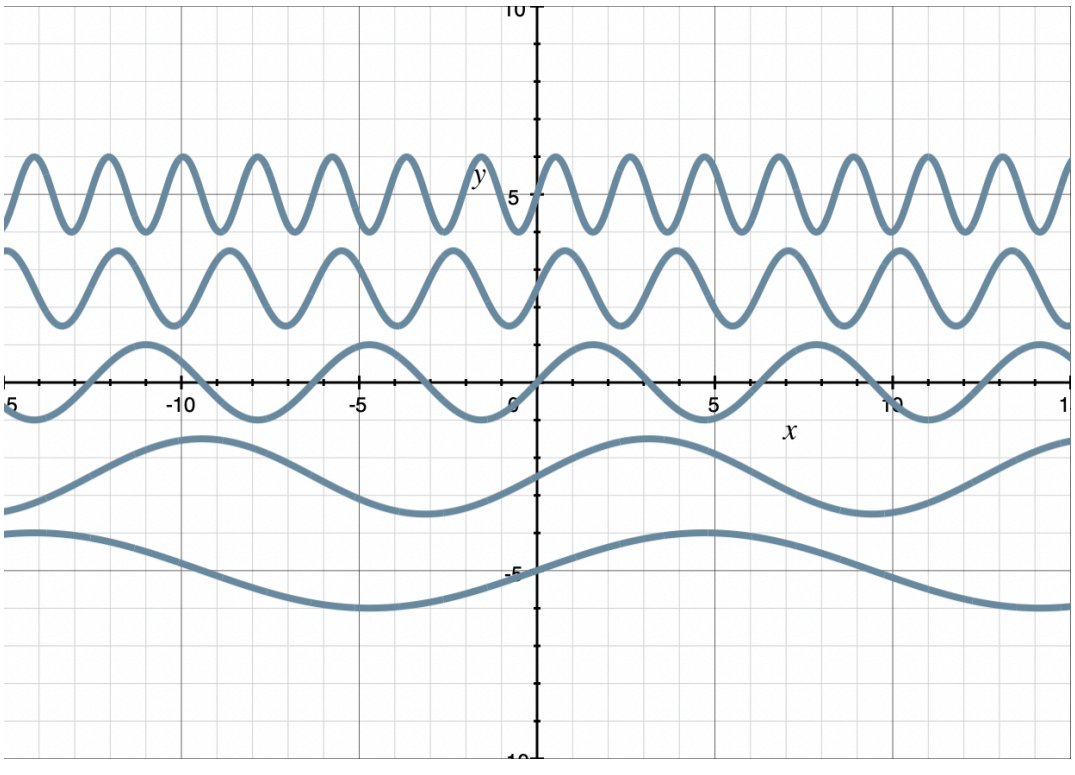
$$f(x) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

Notice that the $n = 0$ term in the first series gives $A_0 \cos(0) = A_0$, so we could also choose to write this as

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

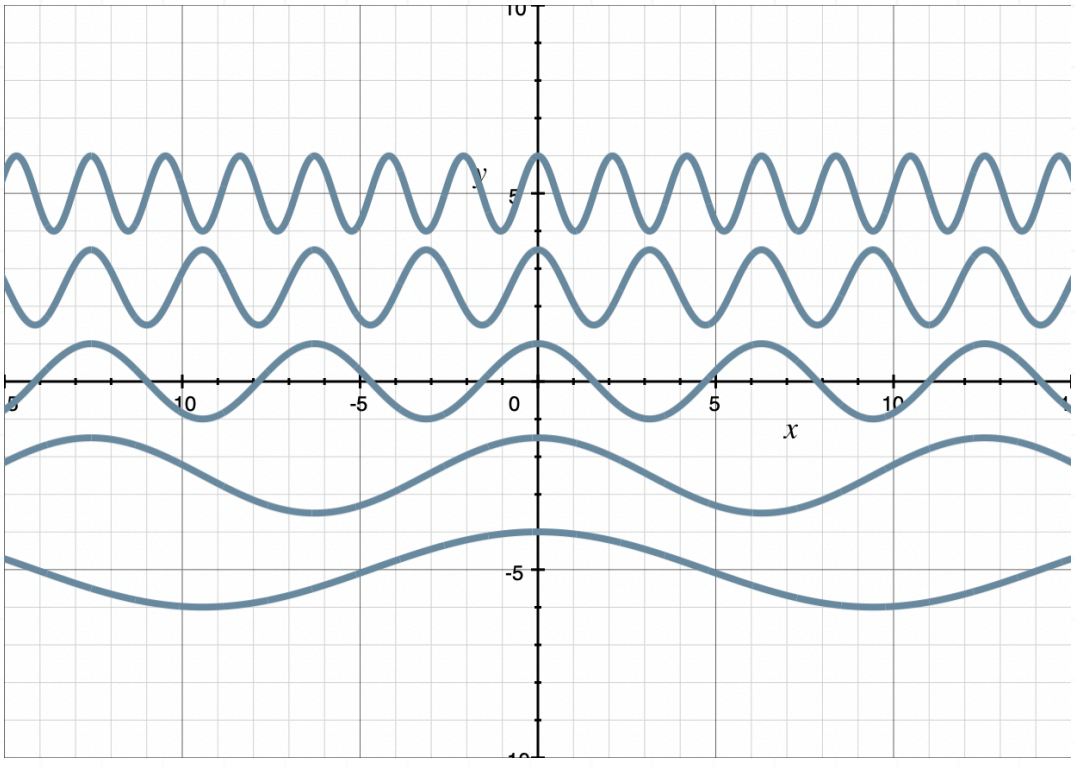
This formula gives us a Fourier series representation of the function $f(x)$ on the interval $-L \leq x \leq L$. Notice that we're just adding a bunch of cosine functions with different arguments (the values of n are given by the series index, and L is given by the interval) in the first series, and then doing the same thing with sine functions in the second series.

This is remarkable! What we're saying here is that, at least in theory, we can express any $f(x)$ as no more than the sum of sine and cosine functions. As a reminder, here are a bunch of sine curves,

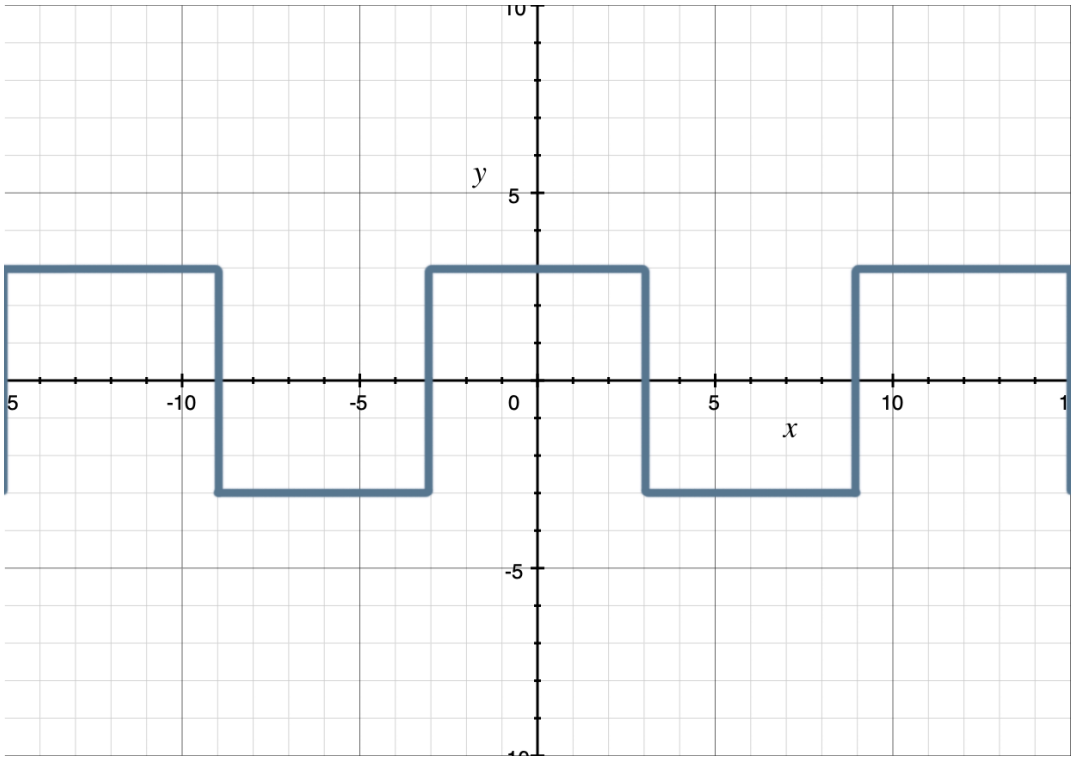


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and here are a bunch of cosine curves,



And now we're saying that if we add up enough of these wavy functions, we can eventually get to the graph of any $f(x)$, regardless of the shape of $f(x)$! Our only condition is that $f(x)$ be periodic, but that still means that $f(x)$ could be made of a bunch of straight lines,



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and we should still be able to find its Fourier series representation. It's pretty incredible to think that we can model straight lines using only the sum of the wavy sine and cosine functions.

That being said, depending on the periodic function $f(x)$ that we're representing, we'll sometimes need to use an infinite number of sine and cosine waves.

And, just like with other series representations (like a Taylor series representation), if we're using only a partial sum to approximate the function (just the first three terms of the series, or just the first five terms of the series, etc.), the more terms we use the more accurate our approximation will be.

But returning now to the Fourier series formula itself, we still have to address A_0 and the coefficients A_n and B_n. They have their own values, which we can calculate using

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\ dx$$

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx \qquad n = 1,\ 2,\ 3,\ \ldots$$

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\ dx \qquad n = 1,\ 2,\ 3,\ \ldots$$

Once we find these values for A_n and B_n, we'll plug them into our Fourier series formula, then simplify as much as we can to get the Fourier series representation of our function $f(x)$.

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Because n is defined for $n = 1, 2, 3, \ldots$ for A_n and B_n, it'll often be the case that we'll find expressions for A_n and B_n that include $\sin(n\pi)$ and $\cos(n\pi)$. For $n = 1, 2, 3, \ldots$, remember that

$$\sin(n\pi) = 0 \qquad n = 1, 2, 3, \ldots$$

$$\cos(n\pi) = (-1)^n \qquad n = 1, 2, 3, \ldots$$

There's also a good chance that we'll use the even-odd trigonometric identities

$$\sin(-x) = -\sin x$$

$$\cos(-x) = \cos x$$

With all of that in mind, let's do a simple example where we just use these formulas to build the Fourier series representation of a function.

Example

Find the Fourier series representation of $f(x) = x$ on $-L \leq x \leq L$.

Let's first find A_0 and the coefficients A_n and B_n. For A_0, we get

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\, dx$$

$$A_0 = \frac{1}{2L}\int_{-L}^{L} x\, dx$$

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$$A_0 = \frac{1}{2L}\left(\frac{1}{2}x^2\right)\Big|_{-L}^{L}$$

$$A_0 = \frac{1}{2L}\left(\frac{1}{2}L^2 - \frac{1}{2}(-L)^2\right)$$

$$A_0 = \frac{1}{2L}\left(\frac{1}{2}L^2 - \frac{1}{2}L^2\right)$$

$$A_0 = \frac{1}{2L}(0)$$

$$A_0 = 0$$

For A_n, we get

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{1}{L}\int_{-L}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{1}{L}\left[x\left(\frac{L}{n\pi}\right)\sin\left(\frac{n\pi x}{L}\right) - \int\left(\frac{L}{n\pi}\right)\sin\left(\frac{n\pi x}{L}\right)\,dx\right]\Big|_{-L}^{L}$$

$$A_n = \frac{1}{L}\left[\left(\frac{Lx}{n\pi}\right)\sin\left(\frac{n\pi x}{L}\right) + \left(\frac{L}{n\pi}\right)^2\cos\left(\frac{n\pi x}{L}\right)\right]\Big|_{-L}^{L}$$

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$$A_n = \frac{1}{L}\left[\left(\frac{L^2}{n\pi}\right)\sin(n\pi) + \left(\frac{L}{n\pi}\right)^2\cos(n\pi) + \left(\frac{L^2}{n\pi}\right)\sin(-n\pi) - \left(\frac{L}{n\pi}\right)^2\cos(-n\pi)\right]$$

Using the even-odd identities we mentioned earlier, we can rewrite this as

$$A_n = \frac{1}{L}\left[\left(\frac{L^2}{n\pi}\right)\sin(n\pi) + \left(\frac{L}{n\pi}\right)^2\cos(n\pi) - \left(\frac{L^2}{n\pi}\right)\sin(n\pi) - \left(\frac{L}{n\pi}\right)^2\cos(n\pi)\right]$$

$$A_n = \frac{1}{L}(0)$$

$$A_n = 0$$

And for B_n, we get

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{1}{L}\int_{-L}^{L} x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = \frac{1}{L}\left[-\left(\frac{Lx}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right) - \int\left(-\frac{L}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_{-L}^{L}$$

$$B_n = \frac{1}{L}\left[-\left(\frac{Lx}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right) + \left(\frac{L}{n\pi}\right)^2\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_{-L}^{L}$$

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$$B_n = \frac{1}{L}\left[-\left(\frac{L^2}{n\pi}\right)\cos(n\pi) + \left(\frac{L}{n\pi}\right)^2 \sin(n\pi) - \left(\frac{L^2}{n\pi}\right)\cos(-n\pi) - \left(\frac{L}{n\pi}\right)^2 \sin(-n\pi)\right]$$

Using the even-odd identities we mentioned earlier, we can rewrite this as

$$B_n = \frac{1}{L}\left[-\left(\frac{L^2}{n\pi}\right)\cos(n\pi) + \left(\frac{L}{n\pi}\right)^2 \sin(n\pi) - \left(\frac{L^2}{n\pi}\right)\cos(n\pi) + \left(\frac{L}{n\pi}\right)^2 \sin(n\pi)\right]$$

$$B_n = \frac{1}{L}\left[2\left(\frac{L}{n\pi}\right)^2 \sin(n\pi) - 2\left(\frac{L^2}{n\pi}\right)\cos(n\pi)\right]$$

$$B_n = \frac{2L}{n\pi}\left[\left(\frac{1}{n\pi}\right)\sin(n\pi) - \cos(n\pi)\right]$$

For $n = 1, 2, 3, \ldots$, $\sin(n\pi) = 0$, and $\cos(n\pi) = (-1)^n$, so the expression for B_n simplifies to

$$B_n = -\frac{2L}{n\pi}\cos(n\pi)$$

$$B_n = -\frac{2L(-1)^n}{n\pi}$$

$$B_n = \frac{(-1)^{n+1}2L}{n\pi} \qquad n = 1, 2, 3, \ldots$$

Then the Fourier series for $f(x) = x$ on $-L \leq x \leq L$ is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

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$$f(x) = 0 + \sum_{n=1}^{\infty} (0)\cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}2L}{n\pi} \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}2L}{n\pi} \sin\left(\frac{n\pi x}{L}\right)$$

Even/odd rules for the coefficients

Before we move forward, let's make a comment here about the integrals of even and odd functions over symmetric intervals.

We may remember from calculus that the integral of an odd function over a symmetric interval is 0, while the integral of an even function over a symmetric interval is equivalent to two times the same integral over half of the interval. In other words, using the symmetric interval $-L \leq x \leq L$ from the Fourier series representation,

$$\int_{-L}^{L} f(x)\, dx = 0 \qquad \text{when } f(x) \text{ is odd } (f(-x) = -f(x))$$

$$\int_{-L}^{L} f(x)\, dx = 2\int_{0}^{L} f(x)\, dx \qquad \text{when } f(x) \text{ is even } (f(-x) = f(x))$$

So when it comes to the integral formulas that we use to calculate A_0, A_n, and B_n,

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\, dx$$

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$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx \qquad n = 1, 2, 3, \ldots$$

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx \qquad n = 1, 2, 3, \ldots$$

because each of them is integrating over the symmetric interval $-L \leq x \leq L$, we can say that

$$A_0 = 0 \qquad \text{if } f(x) \text{ is odd}$$

$$A_0 = \frac{1}{L}\int_{0}^{L} f(x)\,dx \qquad \text{if } f(x) \text{ is even}$$

Because the cosine function in the A_n formula is an even function, the product $f(x)\cos(n\pi x/L)$ will be even when $f(x)$ is even (because the product of two functions is itself an even function), and the product $f(x)\cos(n\pi x/L)$ will be odd when $f(x)$ is odd (because the product of an even function and an odd function is itself an odd function).

$$A_n = 0 \qquad \text{if } f(x) \text{ is odd, } n = 1, 2, 3, \ldots$$

$$A_n = \frac{2}{L}\int_{0}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx \qquad \text{if } f(x) \text{ is even, } n = 1, 2, 3, \ldots$$

And because the sine function in the B_n formula is an odd function, the product $f(x)\sin(n\pi x/L)$ will be odd when $f(x)$ is even (because the product of an even function and an odd function is itself an odd function), and the product $f(x)\sin(n\pi x/L)$ will be even when $f(x)$ is odd (because the product of two odd functions is itself an even function).

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$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx \qquad \text{if } f(x) \text{ is odd, } n = 1, 2, 3, \ldots$$

$$B_n = 0 \qquad \text{if } f(x) \text{ is even, } n = 1, 2, 3, \ldots$$

Looking back at the last example, when we were calculating A_0 we found

$$A_0 = \frac{1}{2L}\int_{-L}^L x\,\,dx$$

which is the integral of an odd function ($f(x) = x$ is an odd function) over a symmetric interval. So instead of doing all the extra work to integrate and evaluate over $[-L, L]$, we could have known right away that $A_0 = 0$.

Similarly, in the last example when we were calculating A_n we found

$$A_n = \frac{1}{L}\int_{-L}^L x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

which is the integral of an odd function ($f(x) = x$ is odd, the cosine function is even, and the product of an even and odd function is itself an odd function) over a symmetric interval. So instead of doing all the extra work to integrate with integration by parts and then evaluate over $[-L, L]$, we could have known right away that $A_n = 0$.

Of course, not every function $f(x)$ will be even or odd; it's actually most likely that the function will be neither even nor odd. But because these even/odd rules can save us so much time in the case when $f(x)$ is even or odd, let's make sure to keep them in mind as we move forward with more Fourier series examples.

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